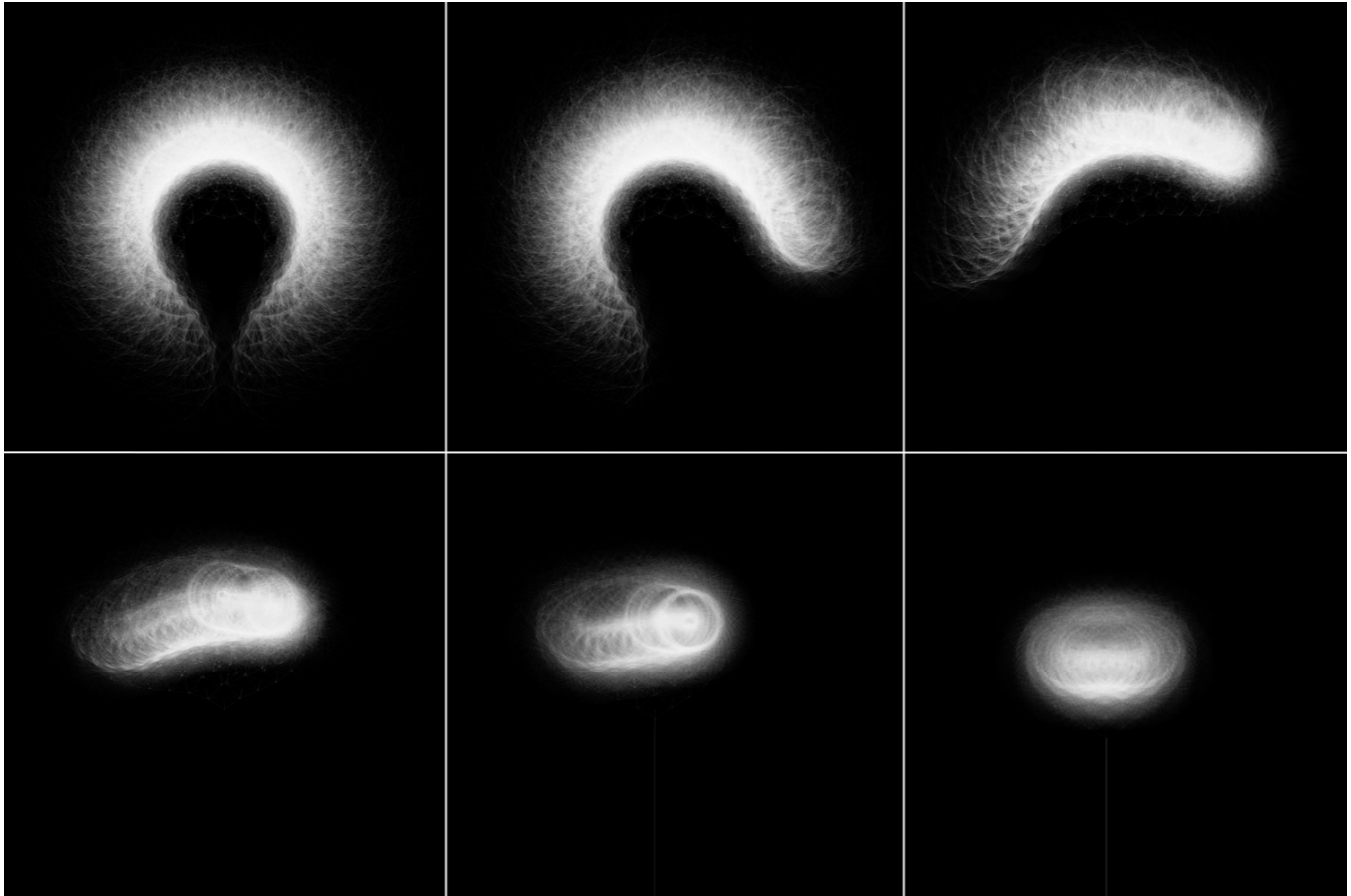


Sensitive Floral

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ABSTRACT

"Sensitive Floral" is an interactive, generative artwork that ventures into the exploration of a unique generative system, biomimetically emulating the reactive behaviors of the Mimosa plant. By synthesizing the complexity of fractal tree data structures with the Cellular Automata mechanisms of grid computations, the artwork elegantly mirrors nature's adaptive and responsive traits. The interactive interface allows users to initiate a ripple of movement by simply touching the screen, triggering a cascade of changes across thousands of leaves, akin to the group leaf movements observed in

Mimosa. The system, instantaneously detects which branch of the tree structure on the touch screen is being externally triggered. It then notifies the grid calculation system, based on the Cellular Automata mechanism, to generate parameters for angle gradients of closing behavior between branches. These parameters are applied in real time to the image generation system of the floral structure, altering the overall appearance of the flower.

The flower shaping heavily employs the recursion mathematical mechanism. Through a set of geometric relationship designs, a concept like cell division is used to grow the next branch. In the iteration process, two characteristics emerge: (1) the development of an organizational relationship of length and angle between lines that presents an organic gradient sense, and (2) the development of movable joints between lines that can display activity for subsequent imitation of the *Mimosa pudica* motion cell mechanism, demonstrating dynamic deformations "Close".

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In existing research, it inspired that the pulvinar cells at the base of Mimosa leaves propagate electrical and chemical signals in response to external stimuli, leading to a chain reaction of leaf movement in adjacent cells. To implement this interlinking characteristic, I employ the computational mechanism of Cellular Automata, both

for aesthetic representation and user sensation, within the tree structure's relationships.

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